Q12 Deeper Project Suggestions

1. **Popularity audit:** take an open model and PopQA-style questions; measure your own frequency–accuracy curve using a public n-gram or crawl-frequency proxy; compare your curve's inflection point with Kandpal et al.'s reported regime.
2. **Adaptive retrieval policy:** implement entity-rarity-gated retrieval (rare $\rightarrow$ retrieve, popular $\rightarrow$ answer directly) and benchmark cost/accuracy against always/never retrieve on a QA set you construct.
3. **Cutoff probe:** build a paired evaluation of pre- vs. post-cutoff facts and skills (e.g., library APIs that changed, events after the date); quantify the fact/skill asymmetry for a specific open model.
4. **Abstention training:** fine-tune a small model with explicit "I don't know" targets on unanswerable questions and measure the trade-off between abstention rate and coverage on answerable ones.